Tribhuvan University

Faculty of Humanities and Social Sciences

“Job Portal” Using Content Based Filtering and Rule Based Matching for Recommendation and Decision Tree for Salary Expectation

A PROJECT REPORT

Submitted to

Department of Computer Application

Asian College of Higher Studies

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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11, October 2024

Under the Supervision of

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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by Nhujan Maharjan entitled "**Job Portal Using Content Based Filtering and Rule Based Matching for Recommendation and Decision Tree for Salary Expectation**" in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

…………………………………………………………………………

Janak Kumar Lal

SUPERVISOR

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LETTER OF APPROVAL

This is to certify that this project prepared by Nhujan Maharjan entitled "Job Portal" Using Cosine Similarity And Name Entity Relation For Recommendation And Decision Tree for Salary Expectation in partial fulfillment of the requirements for the degree of Bachelor in Computer Applicationhas been evaluated. In our opinion, it is satisfactory in the scope and quality as a project forthe required degree.

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-- **Mr. Janak Kumar Lal** **Asian College of Higher Studies**	-- **Mr. Pranaya Nakarmi** **Asian College of Higher Studies**
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-- **Mr. …………………….** **Internal Examiner**	-- **Mr. …………………..** **External Examiner**

Abstract

Searching for a job can be a challenging and time-consuming process, especially when relying on traditional methods like scanning newspaper job posts or seeking referrals from friends and relatives. These conventional approaches often yield limited results, making it difficult to find a job that aligns with one's skills and preferences. This is precisely where the invaluable role of a job portal comes into play. To address these challenges and enhance the job-seeking experience, we have developed a system that leverages skill-based job recommendations. Our system provides an easy-to-use website where job seekers can enter their details. This helps them get personalized job suggestions that match their skills and career goals. Additionally, our system helps with another common concern job seekers have: figuring out how much they might earn based on their skills and experience. For which our system offers the valuable feature of predicting salary based on a job seeker's skills and experience, employing a decision tree algorithm. Similarly, another motive of this project is to find out the best candidate for the announced vacancy and recommend the perfect candidate to the employer. Main goal of this system is to make it easier and less stressful for people to find jobs by providing job seekers with the tools and information they need to successfully navigate the job market.

Keywords: Job Portal, Content-based filtering, Cosine similarity, Decision tree.

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Project By

Nhujan Maharjan

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List of Abbreviations

CSS	Cascading Style Sheet
CV	Curriculum Vitae
ER	Entity Relational
GB	Giga Byte
HTML	Hypertext Markup Language
PHP	Hypertext Preprocessor/ Personal Home Page
RAM	Random Access Memory
SSD	Solid State Drive
SQL	Structured Query Language
UI	User Interface
VS	Visual Studio

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Chapter 1: Introduction to Job Portal

1.1 Introduction

In today's competitive job market, where numerous candidates compete for a limited number of positions, finding the right job can be quite challenging. Job portals have become essential tools, serving as a bridge between job seekers and employers by efficiently matching individuals with positions that fit their skills and qualifications. These platforms simplify the job search process by consolidating job listings in one convenient place, allowing candidates to save time and effort. Instead of scouring multiple websites, job seekers can easily access a wide array of opportunities from different industries and locations, all in one spot.

The growing reliance on job portals has revolutionized the traditional job search process, making it more accessible and convenient for job seekers and employers alike. Many job portals now offer personalized recommendations based on user profiles, providing tailored job suggestions that match the skills, experience, and career aspirations of the applicant. This added layer of intelligence has greatly improved the chances of landing a relevant job, as candidates can focus their efforts on opportunities that best fit their qualifications.

Job portals not only help candidates explore diverse roles that align with their career goals but also offer valuable insights into market trends, company profiles, and industry demands. This added information allows job seekers to make informed decisions and prepare for the application process more effectively. On the other side, employers' benefit from these platforms by gaining access to a vast pool of potential candidates, enabling them to identify individuals whose qualifications and experiences are tailored to their specific needs. For employers, job portals provide sophisticated tools to filter and sort applications, making the recruitment process more efficient.

In essence, job portals create a streamlined, user-friendly environment where job seekers can effortlessly search for roles that match their qualifications, and employers can efficiently connect with the right talent. This makes the recruitment process smoother and more productive for everyone involved, contributing to a more dynamic and responsive job market. The study finds 78% of seasoned job seekers used the internet to search for employment opportunities, versus only 64% of recent graduates. More experience job-seekers were alsomore likely to register on more than one job portal and relied less on social networks (friends, immediate family, classmates, relatives, etc.) than recent graduates [1]. The use of job portals to find employment has significantly increased in recent years. People register on these platforms to search for positions that match their skills and qualifications, making the job-hunting process much more efficient. Even those already employed often turn to job portals in search of better career opportunities. Job portals serve as a crucial link between job seekers and employers, helping to match the right candidates with the most suitable job openings.

These platforms cover a wide range of job categories by aggregating listings from various company websites or by receiving job postings directly from employers. This streamlined approach makes it easier for job seekers to browse and apply for roles that fit their qualifications. Moreover, job portals facilitate direct connections between job seekers and hiring companies, simplifying the communication and recruitment process. This collaboration between job seekers and employers is reshaping the way people find and secure jobs, offering greater convenience and efficiency for both sides.

Job portals have become indispensable in connecting job seekers with the right opportunities while giving employers a platform to efficiently discover talent. With their ability to offer personalized job recommendations, streamline recruitment processes, and enhance the overall job search experience, these platforms are shaping the future of employment and ensuring a smoother journey for both candidates and hiring managers.

1.2 Problem statement

Searching for jobs through newspapers is a slow and frustrating process, often making people wait for months to find a job that fits their skills and qualifications. Many online platforms that offer job opportunities have outdated listings, which makes it hard for job seekers to find up-to-date openings. On top of that, fake employers and misleading job ads add more challenges and risks to the online job search.

While many websites allow people to search for jobs, they often don't recommend roles that match the seeker's skills, qualifications, or preferences, making the process even harder. Job seekers may spend hours scrolling through listings that are not relevant to them, leading to feelings of discouragement and burnout. This situation not only wastes valuable time but also hinders individuals from finding meaningful employment that aligns with their career goals. As a result, job seekers can feel lost and overwhelmed in their search, and many may miss out on opportunities that are a better fit for their talents and aspirations.

To address these issues, it is crucial to have a more efficient and reliable system that connects job seekers with genuine employers and offers tailored job recommendations, helping to streamline the search process and improve the chances of finding suitable employment.

1.3 Objectives

The objectives of the system are:

a) To build a Web Application that delivers tailored job suggestions for job seekers and recommend suitable candidates to employers based on specific requirements.
b) To utilize a decision tree algorithm to predict salaries based on skills and experience, and enable direct applications for a more convenient job search experience.

1.4 Scope and Limitation

Scope are as follows:

- Job Seeker can view the vacancies listed by different employers.
- Job Seeker can post their resume.
- Job Seeker can manage and update their resume.
- Job Seeker can apply for the vacancy post.
- Employer can view the application.
- Employer can manage their vacancies.

- Employers can list their job vacancy on the site.

Limitations are as follows:

- This system does not provide any personalized support.
- Vague job descriptions, limited information provided by job seekers can lead to mismatch recommendation.
- Listing of outdated or non-existing job vacancy.

1.5 Development Methodology

The Waterfall methodology is a linear, sequential approach to project management, where each phase must be completed before the next one begins. The progress of the project is easier to track due to its linear nature, allowing for clear milestones, deadlines, and checkpoints to ensure the project stays on schedule. Once requirements are set at the beginning, scope changes are minimal. This reduces the risk of unexpected alterations, ensuring that the project stays focused on its original goals.

So, we chose this methodology for our project because it provides a clear structure and ensures that all requirements are thoroughly defined before development starts. This helps minimize risks and scope changes later in the project, making it ideal for projects with well-understood requirements and a fixed timeline. The focus on detailed documentation at each stage ensures better tracking and accountability throughout the project lifecycle.

Additionally, Waterfall's structured phases gives us the ability to thoroughly test and verify each component before moving on to the next, reducing the chances of major issues emerging late in the development process. This approach offers predictability and stability, which are critical for the successful and timely completion of our project.

1.6 Report Organization

Chapter 1:

This chapter includes the introduction of the project where problem statement, objective of the project along with the scopes and limitations of the project are listed. This will provide a general concept of the project to the reader. They can alsoable to know how the system can work and for what purpose it was developed.

Chapter 2:

It includes background study and literature review of a similar project and conceptsof the current project. A literature review involves gathering information from works published by other authors on the same or related subjects. In this instance, books and papers were examined to gain insights into various topics.

Chapter 3:

It contains the workings of the project including feasibility analysis, functional andnon-functional analysis along with the schema and architectural design and the data and process modeling diagrams.

Chapter 4:

It includes the testing and implementation phase of the project i.e., designing of testcases to check if the system and its components works as intended during development. It also contains tools that were used in this project.

Chapter 5:

It contains the conclusion of the entire project. In this chapter, outcome of the project is written along with the things learnt from the development of the project.The future recommendations of the project are also listed in the end of chapter 5.

Chapter 2: Background Study and Literature Review

2.1 Background Study

Historically, searching for job advertisements in newspapers was the traditional way for job seekers. Individuals would look through the printed pages, marking positions that caught their interest. However, this method had several challenges. The task of sifting through dense text to find suitable job opportunities can be compared to trying to find a small object in a large haystack; there is a significant risk of missing out on promising job openings simply because they are on a different page or in another section of the paper.

Another option is to use job agencies, which often required payment for their services, but this approach did not always produce the desired results. The current job market is highly competitive, making it increasingly difficult for both job seekers and employers. Companies may struggle to find the right candidates, while job seekers may not be aware of available job openings.

Fortunately, advancements in technology have greatly improved this situation. The internet and mobile devices have created closer connections between job seekers and employers. Today, job seekers can access a wide range of job opportunities through online platforms. Some websites even provide personalized job recommendations based on users' preferences and qualifications. However, the accuracy of these suggestions can be limited, as many platforms rely on inadequate user data. Additionally, some job portals focus more on quickly matching jobs rather than ensuring that candidates are well-suited for positions based on their skills and qualifications.

2.2 Literature Review

Based on our research findings, a significant majority of job portal systems uses collaborative filtering algorithm and content-based filtering algorithm for job recommendations.

Collaborative filtering is a widely used recommendation technique that relies on the behavior and preferences of users to predict their future preferences. This approach is grounded in the principle that users who have similar tastes in the past will continue to share preferences in the future. Collaborative Recommendation technique gives recommendation based on items and target user and consider past history of similar users with similar choices [2].

The collaborative filtering method is based on the historical interlinkages that are documented among the users and items. The method tries to forecast the usefulness of articles for a specific user on the articles previously evaluated by fellow users. These memory-based methods work directly with the recorded interaction values and are based on finding the nearest neighbor, for example, finding the closest user of interest and suggesting the most popular product [3].

One of the biggest disadvantages of memory-based collaborative filtering is the fact that, in order to work, system must be fed with the entire dataset. This becomes a significant issue when a user-item correlation matrix becomes too large for a big number of users. It can have a lot of influence to system’s performance by consuming more computational power than it should. As a result, the system cannot respond to user’s request in desired time [4].

Jeevamol Joy and Renumol V G discussed which similarity is the best one for a content-based recommending system. They finally concluded that cosine similarity is the best similarity for content based recommended system. Cosine similarity is not only used for recommender systems but is used to find the similarity functions between two sentences or two paragraphs [5].

Content-based filtering is an algorithm that suggests things (like jobs) to users based on their interests and preferences. It looks at the details of items (like skills required for a job) and matches them with what the user likes. If the item's details match the user's preferences, it's recommended to them. This approach focuses on using the content of items and the user's past choices to make personalized suggestions.

Content-based recommendation systems cluster users by comparing representations of items to representations of users’ profiles. Representations of users’ profiles express users’ interests. LinkedIn’s system was at first based on content-based approach. After learning the advantages of hidden fields in users’ profiles, LinkedIn decided to replace user profile-based fields with those hidden fields. Then, those fields were used for understanding the taste of each user by creating a ranking model [6].

As content-based filtering brings personalized recommendations by looking at item details and matching them with your preferences. This approach is useful when you have unique tastes or limited data. It also helps you discover new options aligned with your interests and provides transparent suggestions based on item attributes and your preferences.

Decision tree regression is a powerful machine learning technique used to predict salary in this system. It works by breaking down the data into a tree-like structure of decisions and outcomes. In the context of salary prediction, it considers factors like years of experience, and job role to make informed salary estimations.

Decision trees have proved to be valuable tools for the description, classification and generalization of data. Work on constructing decision trees from data exists in multiple disciplines such as statistics, pattern recognition, decision theory, signal processing, machine learning and artificial neural networks. Researchers in these disciplines, sometimes working on quite different problems, identified similar issues and heuristics for decision tree construction [7].

Chapter 3: System Analysis and Design

3.1 System Analysis

3.1.1 Requirement Analysis

i. Functional requirement

Functional requirement basically means what system is required to execute.

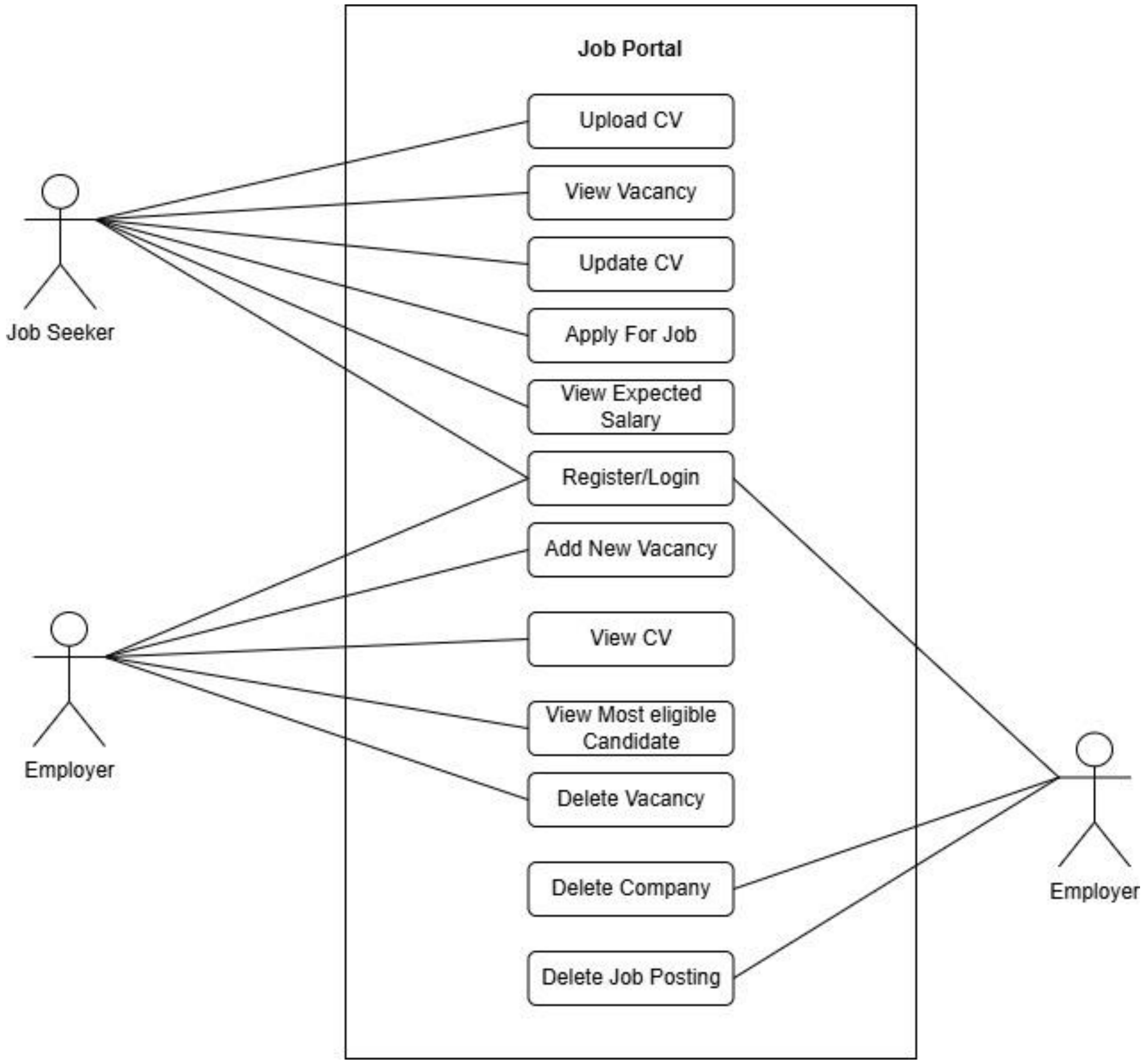

Figure 3.1: Use case diagram for Job Portal

The depicted diagram illustrates the functionalities available to users when utilizing the website. Job seekers can explore job openings from various companies and apply for positions that match their qualifications. They have the option to upload and refresh their CVs. Employers or companies can post new job vacancies and review the CVs of job seekers. Additionally, administrators possess the capability to monitor employers, verify the authenticity of companies or employees, and remove employers.

ii. Non-Functional Requirement

a. It is easy to use, efficient, and accessible.
b. This system is available 24*7 to the user.
c. Any modification like insert, delete, update, etc. for the database can be synchronized quickly and can only be executed by the administrator.
d. This system should support all browsers like: google chrome, Firefox, safari, etc.
e. The website should be secure, protecting both the personal information.

3.1.2 Feasibility study

Feasibility study is an evaluation and analysis to determine if the project is viable and practical. Some major feasibility studies of this project are: -

i. Technical feasibility

The proposed system is technically feasible as we are going to develop the system using existing technology. The required hardware and software for the development of the systemare available. The software will be developed using Markup Language, Stylesheet and programming languages like HTML, CSS, and JavaScript for front end and PHP will be used as the back end for this project. With every knowledge of working with programminglanguages, the system will be developed. Hardware requirement for the project is desktop or laptop having 4GB RAM or more, Intel core i5 5^{th} gen, 512GB SSD. Software like Wamp Server, MySQL, Windows 10, Vscode for Development environment will be used.

ii. Operational feasibility

The system will be easy to operate. It does not require high level of technical knowledge to use the system. People having basic knowledge in computer operating can use the system very efficiently as it is easy to use and does not require advanced technical skills. People with basic computer knowledge will be able to operate it without any trouble. The system is designed to be straightforward, allowing users to complete tasks quickly without complicated steps.

Clear instructions and helpful error messages are provided, making it easy for users to resolve issues if they occur. Many tasks are automated, reducing the need for users to do everything manually and helping them save time.

iii Economic feasibility

The cost for the operation of this project is very minimum. As the project will be developedon existing technologies and medium end device the development cost is very low. The operation cost to use the system can vary, as the system might charge for some premium services from the user. Although the benefits that user gains by spending a small amount ofmoney will be comparatively high.

iv. Schedule feasibility

The project will be completed within 5 months. The process to create the system is divided according to have successful completion.

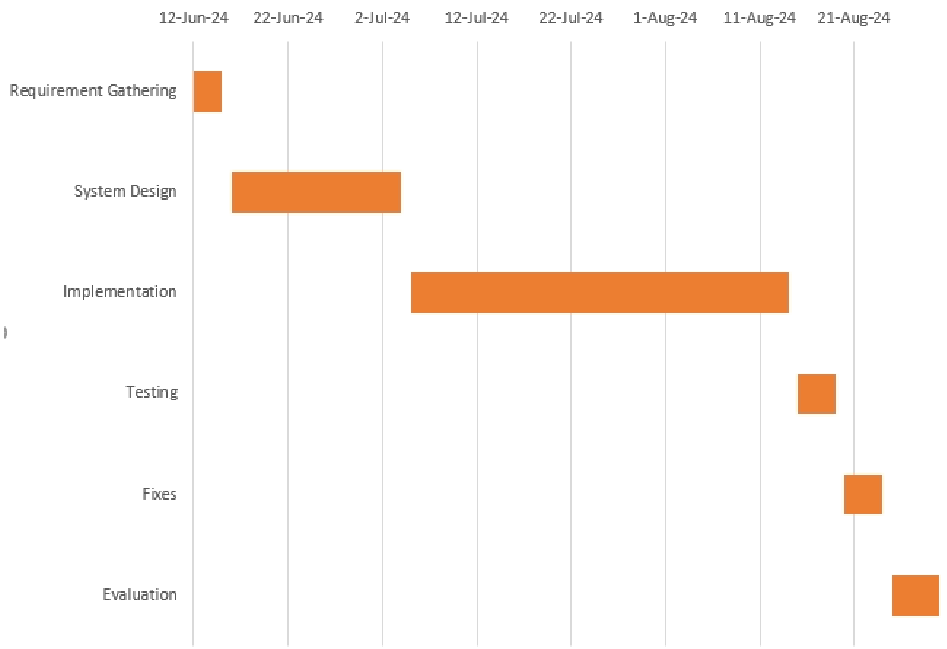

Figure 3.2: Gantt Chart of Project

The project will begin with Requirement Gathering from 12th to 15th June 2024, taking 3 days to gather all necessary details. This will be followed by System Design, lasting from 16th June to 4th July 2024, for a total of 18 days. After that, the Implementation phase will run from 5th July to 14th August 2024, taking 40 days to build the system. Once implementation is complete, the system will undergo Testing from 15th to 19th August 2024, taking 4 days. Any required Fixes will be made from 20th to 24th August 2024, also over 4 days. Finally, the system will go through an Evaluation phase from 25th to 30th August 2024, which will take 5 days to complete.

3.1.3 Object Modeling using Class and Object Diagrams

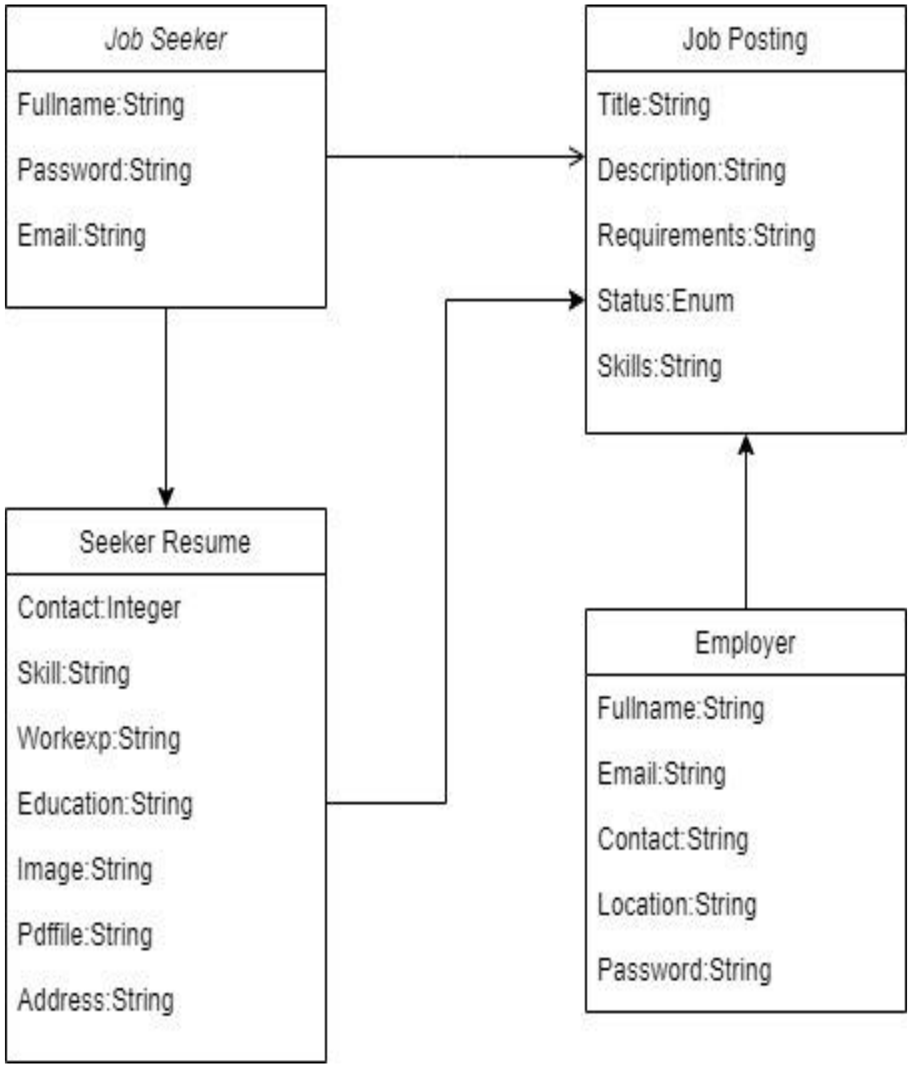

Figure 3.3: Class diagram of Job Portal

A class diagram is a specialized type of UML (Unified Modeling Language) diagram that plays a crucial role in the design and visualization of a system or software application. It serves as a blueprint that outlines the static structure of a system by illustrating the various classes, interfaces, and objects that comprise it.

In a class diagram, each class is represented as a rectangle divided into compartments, showcasing essential information such as the class name, attributes (or properties), and methods (or functions). This organized format allows developers and stakeholders to quickly grasp the roles and functionalities of different components within the system.

3.1.4 Dynamic Modelling using State and Sequence Diagrams

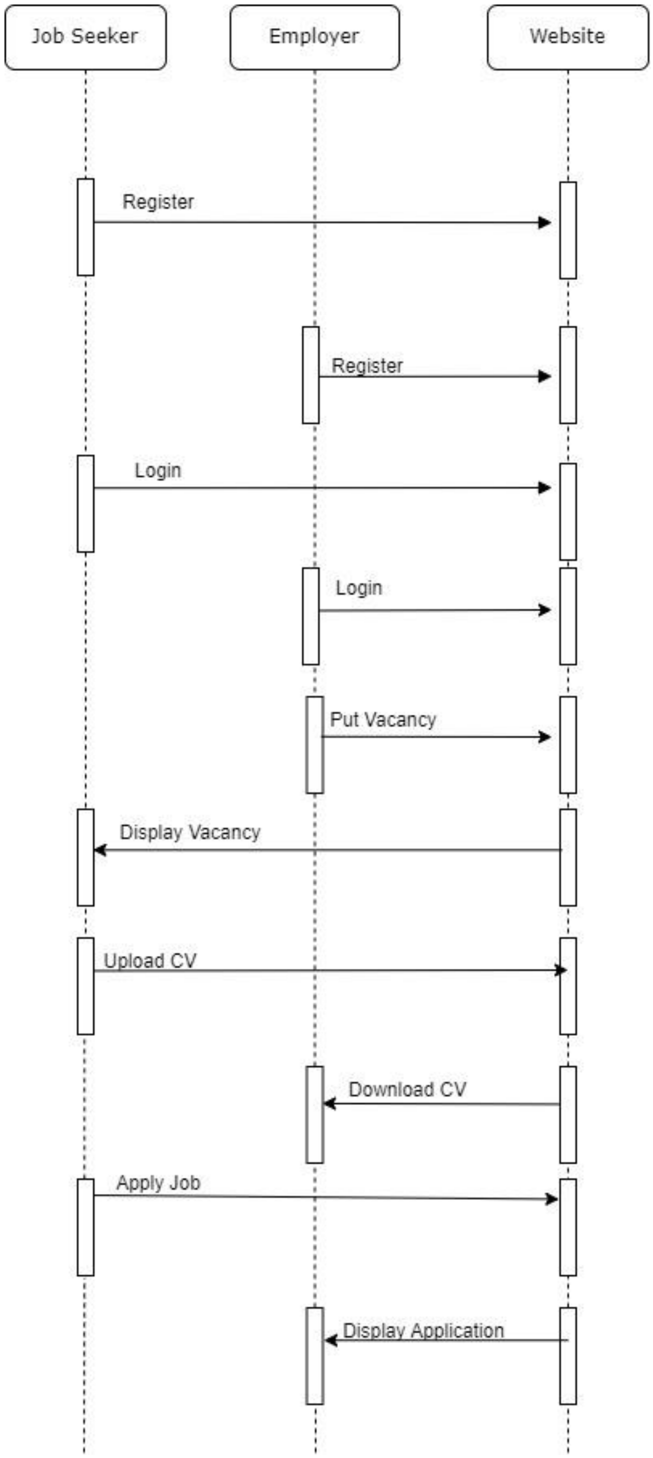

Figure 3.4: State and Sequence diagram of Job Portal

A sequence diagram is type of UML diagram that illustrates how objects interact in a particular sequence over time. It shows the order of messages exchanged between objects, which helps visualize the flow of control in a system. The sequence diagram illustrates the interaction between a user and a job portal system, detailing key activities such as registration, login, CV management, job posting, and job applications. Each step begins with the user initiating an action—like registering or uploading a CV—which the job portal system processes. The system then communicates with the database to store or retrieve information, ensuring data integrity and confirming successful actions back to the user. This visual representation clarifies the flow of information and the sequence of events, making it easier to understand how the different components of the job portal interact. By mapping out these processes, the diagram aids in identifying potential areas for improvement and ensures that all user interactions are streamlined and efficient.

3.1.5 Process Modelling using Activity Diagrams

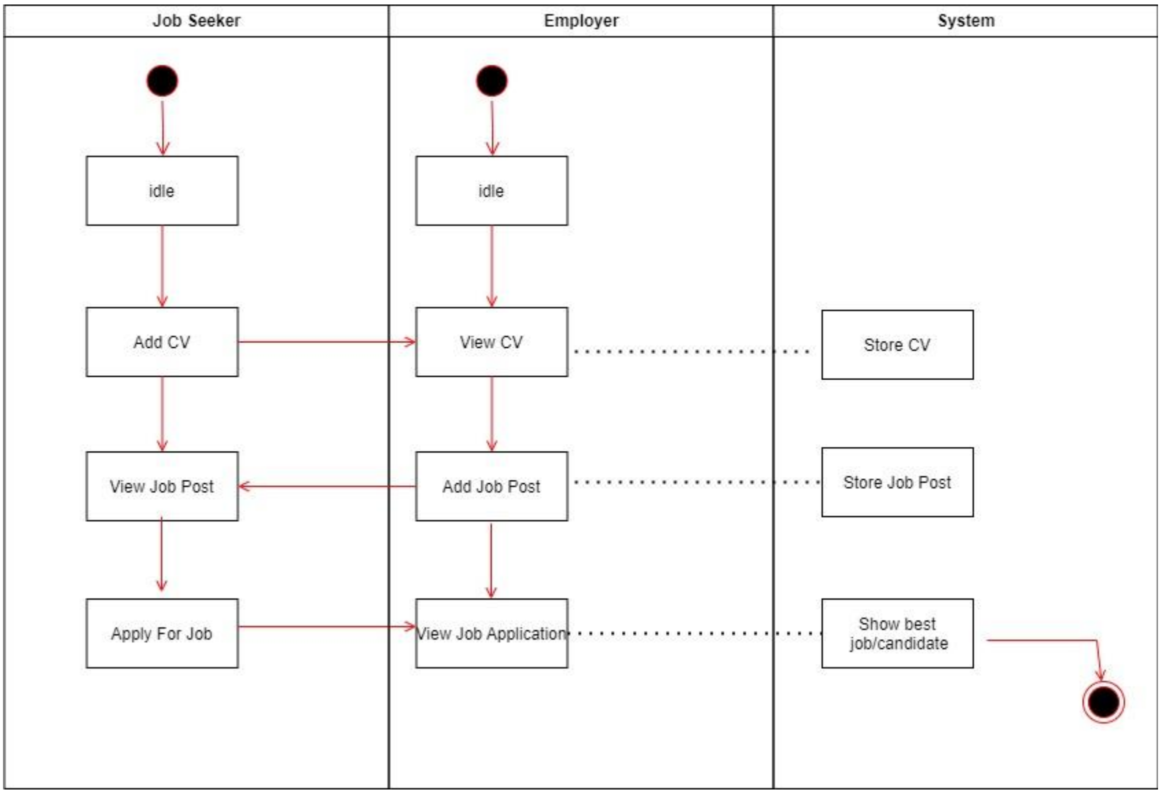

Figure 3.5: Activity diagram of Job Portal

The activity diagram outlines the workflow of a job portal, showing the various activities a user can perform. Each activity is a step in the user journey, with conditions for validation that ensure proper data handling. This detailed view helps in identifying all possible user interactions with the system, ensuring that every aspect of the job portal functionality is covered. By visualizing these processes, the diagram aids in designing the user experience, facilitating system development, and enhancing the overall usability of the job portal.

3.2 System Design

3.2.1 Refinement of Class, Object, State, Sequence and Activity diagrams

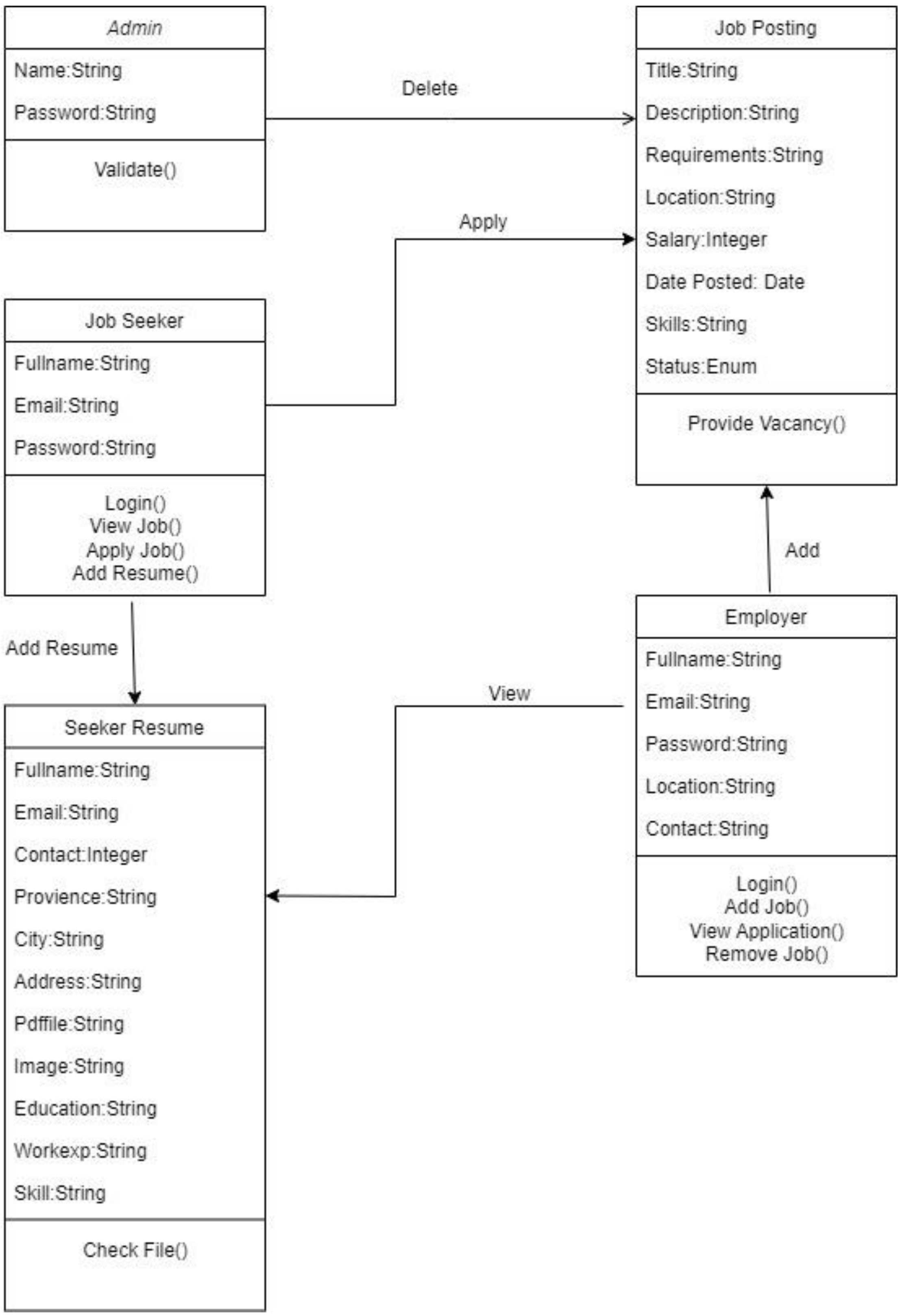

Figure 3.6: Refinement of Class Diagram

This refined class diagram clearly defines the structure of the job portal system. Each class is detailed with its attributes and methods, showcasing the functionality of the system. The relationships indicate how different classes interact with one another, providing insight into data flow and dependencies within the system.

By organizing the components in this way, developers can easily identify key elements of the system, facilitating effective design, implementation, and future enhancements.

3.2.2 Component Diagrams

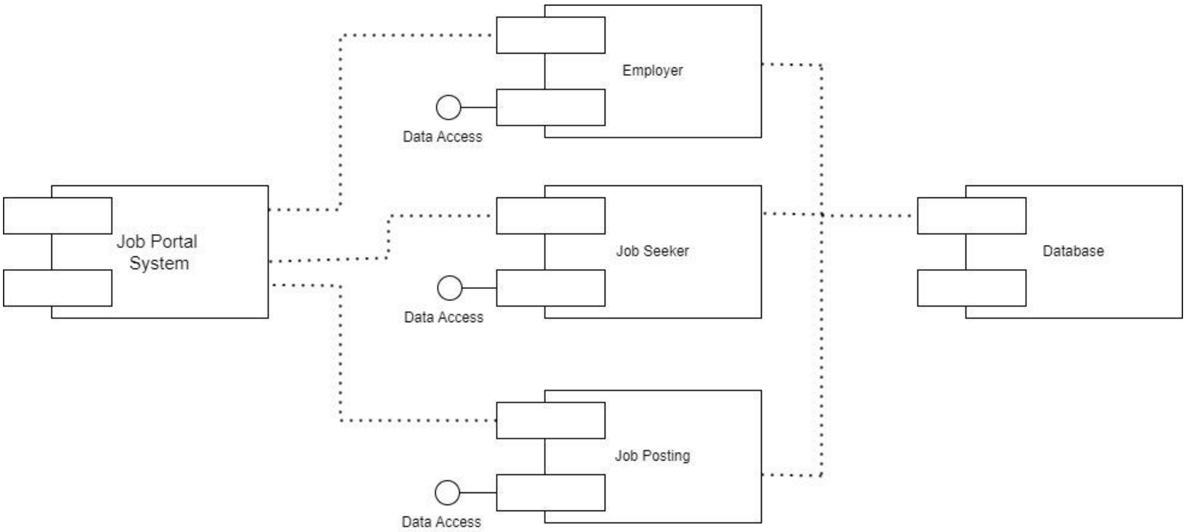

Figure 3.7: Component Diagram of job Portal

The component diagram for the job portal outlines the key elements of the system, highlighting the interactions between employers, job seekers, and job postings. At the core is the Job Portal System, which includes the user interface for both employers and job seekers, along with an application server and a database for data management. The Employer component features a dashboard for managing job postings and applications through a dedicated job posting service. The Job Seeker component provides a dashboard for uploading CVs and applying for jobs, supported by an application service to handle submissions. Additionally, the Job Posting component encapsulates the details of each job listing and an application tracker for managing candidate responses. This diagram illustrates the structure and relationships among components, clarifying the overall functionality of the job portal.

3.2.2 Deployment Diagrams

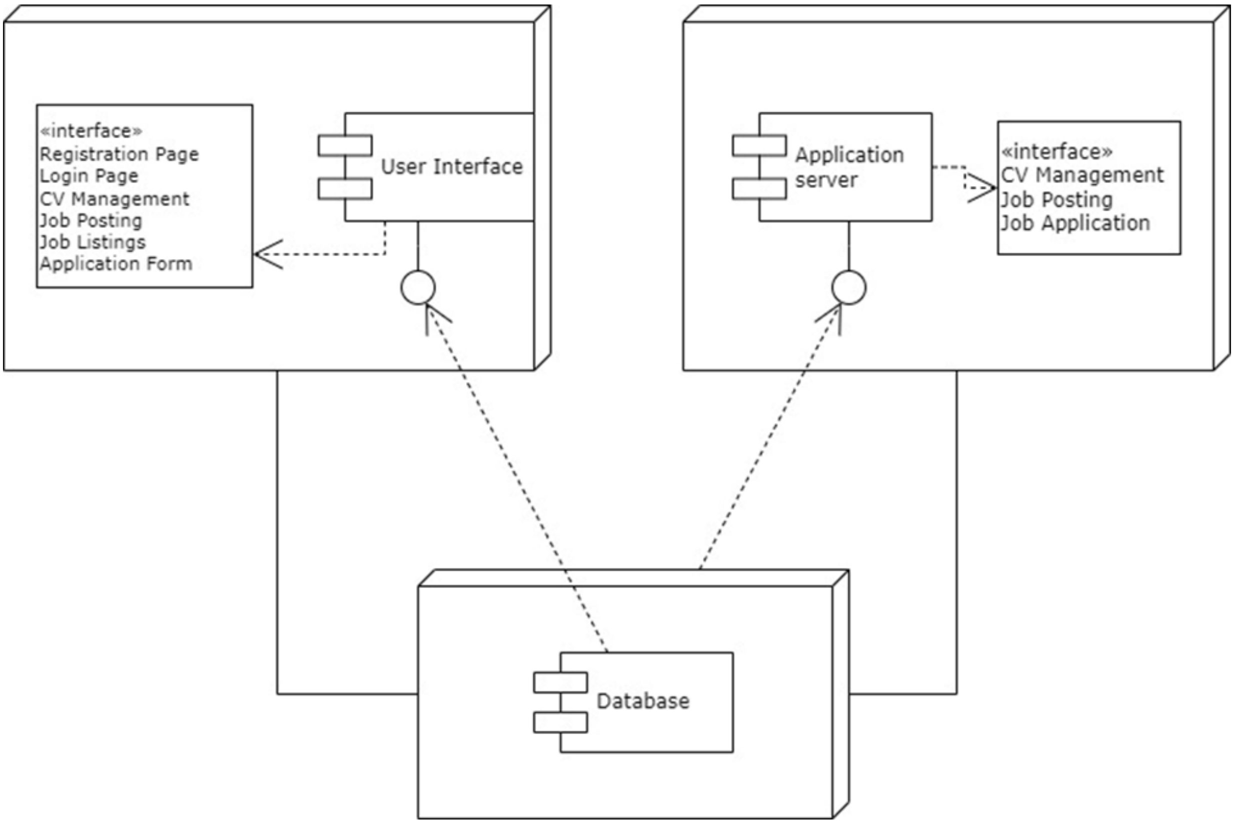

Figure 3.8: Deployment Diagram of Job portal

Above figure shows that the User Interface (UI) of the job portal provides an intuitive platform for users to interact with the system, featuring pages for registration, login, CV management, job postings, and applications. The Application Server acts as the core engine, processing user requests, managing business logic, and coordinating data transactions between the UI and the database. Finally, the Database serves as the secure storage layer, housing user information, CVs, job postings, and applications, ensuring data integrity and efficient retrieval to support the portal's operations. Together, these components create a seamless and functional job portal experience.

3.3 Algorithm details

3.3.1 Content based algorithm

We have implemented a content-based filtering algorithm for recommending jobs, using cosine similarity to match the skills of a user with those required for various job postings stored in a database. In this approach, the algorithm first extracts the skills of both the user and the jobs, treating them as vectors. It then applies cosine similarity to compute the similarity score between the two vectors. Cosine similarity is calculated using the formula:

Cosine Similarity (A, B) = $(A \cdot B) / (\|A\| * \|B\|)$

Where A represents the user's skill vector, B and represents the job's required skill vector. The dot product (A.B) measures how many skills overlap between the user and the job, while the magnitudes ($\| A \|$ and $\| B \|$) normalize the vectors to give a score between 0 and 1. A higher score indicates a closer match.

In the code, the algorithm first retrieves all job postings and their required skills, then calculates the cosine similarity between the user's skills and each job. Jobs with a similarity score greater than zero (i.e., those that share at least one skill with the user) are recommended. The recommendations are sorted in descending order based on the similarity score, ensuring that the most relevant job options are presented first. This content-based approach focuses on the user's attributes, in this case, their skills, to make accurate job recommendations.

By leveraging cosine similarity, the algorithm not only identifies matching jobs but also ranks them based on how closely the job requirements align with the user's skills. This method simplifies the job search process, enabling job seekers to efficiently find opportunities that are most suited to their qualifications.

3.3.2 Decision tree algorithm

We have implemented a decision tree regression algorithm, a machine learning technique aimed at predicting numeric outcomes based on input features. Specifically, it is designed to estimate salaries based on a user's years of experience. The algorithm begins by examining a dataset consisting of pairs of experience and salary values. It uses statistical functions to calculate the variance and mean, which help evaluate potential splits in the dataset.

To assess the quality of a split, the algorithm employs the Mean Squared Error (MSE) formula:

$$MSE = 1/n * \Sigma ((y\_i - \bar{y})^2)$$

where y_i represents the actual salary values and ȳ represents the predicted salary values for the dataset. The goal is to find splits that minimize the MSE, ensuring that the predictions are as accurate as possible.

The decision tree is built recursively, starting with the entire dataset and considering all possible splits based on the input features. Each branch of the tree corresponds to a decision point, such as whether the experience is greater or less than a specific threshold. This splitting continues until certain criteria are met, such as reaching a maximum depth or having too few samples to split further.

Once built, the tree can be used to predict salaries for new inputs by traversing the tree according to the conditions defined at each node. The prediction is based on the mean salary of the subset of samples that reach the terminal node. This algorithm simplifies salary prediction based on experience and can be easily extended to other numeric prediction tasks, making it a powerful tool for estimating outcomes in various scenarios.

3.3.3 Rule Based Matching

The rule-based matching system designed to extract and analyze candidate information from CVs, thereby identifying those who best fit a specific job description. The process begins with text extraction from CV files, utilizing dedicated methods for different formats—PDF and DOCX. For PDF files, a parser retrieves the text, while a library is used for DOCX files to read the document and concatenate text from various sections.

Once the text is extracted, the algorithm employs regular expressions to identify and extract specific entities. Names are extracted using a pattern that matches common titles followed by the first and last name. Email addresses are identified using a standard email format regex, while phone numbers are captured with a pattern that accommodates various international formats. Dates are also extracted through a regex designed to recognize multiple date formats. Additionally, the algorithm checks for the presence of predefined skills—such as PHP, JavaScript, and Python—by scanning the extracted text for these keywords.

To evaluate the relevance of each CV against a given job description, the algorithm tokenizes both the job description and the CV content, allowing for a more accurate comparison of the two. The scoring mechanism calculates a relevance score based on matching tokens, which is derived from the presence of keywords from the job description in the CV. The formula considers matches for desired educational degrees and job titles, ensuring that candidates who possess the necessary qualifications are prioritized.

Overall, this algorithm streamlines the recruitment process by scoring CVs according to predefined rules, enabling employers to swiftly identify the most suitable candidates. By focusing on key attributes such as skills, experience, and educational qualifications, the algorithm effectively highlights the best matches for further consideration, enhancing the efficiency of the hiring process.

Chapter 4: Implementation and testing

4.1 Implementation

This implementation is the second last chapter of this project. It discusses programming language or software used in this application. It also gives details of codes of the entire system. This chapter also shows how the system works as a whole.

4.1.1. Tools used

a) Case tools

- Vs Code:

  VS Code is a streamlined code editor with support for development operations like debugging, task running, and version control. It features a lightning-fast source codeeditor which supports hundreds of languages. We have done all the coding in VS Code. VS Code is very easy to use and provide a developer friendly environment.

- XAMPP:

  XAMPP is a free and open-source cross-platform web server solution stack packagedeveloped by Apache Friends, consisting mainly of the Apache HTTPS server, MariaDB database and interpreters for scripts written in the PHP and Perl programming languages. It is mainly used to host the server for using the website.

- Draw.io:

  Draw.io is a free, open-source windows app that helps to create offline or online diagrams. Many diagrams can be created easily like UML diagrams, flowchart,etc. We have made diagrams using Draw.io.

- Dbdesigner.net:

  Dbdesiner.net is a database schema design and modeling tool. It lets us use databaseschema without using SQL. We have made our database schema using Dbdesigner.io.

b) Programming language

There are many programming languages which are used in this project. Some of thelanguages are HTML, PHP, JAVASCRIPT, etc.

- HTML:

  With the help of html, we created our websites. This language is used as the frontendof the website. Html is used as base of our project. With help of html, we have described the structure of the webpage. The content on the system is established using html.

- CSS:

  CSS used to style the HTML document of our website. CSS have been used to givestyles to our website. With the help of CSS, we have tried to make our website quiteattractive.

- PHP:

  PHP is used for backend. All the data manipulation, update, remove are done usingPHP. As a general scripting language, it was quite easy for us to use. All backend isdone from PHP.

- JAVASCRIPT:

  JavaScript is a lightweight, interpreted, or just-in-time compiled programming language with first-class functions. Using JavaScript, we have done mostly form validations.

c) Database platforms

- PhpMyAdmin:

  We have used for database called MYSQL. MYSQL is an open-source SQL relational database management system that's developed and supported by Oracle.It is a system that stores and manages the data. Using PhpMyAdmin we have madeour database and tables.

4.1.2 Implementation details of modules

Different implementations functions are as follows:

Recommend job through matching skills

```
$sql = "SELECT * FROM seekerresume WHERE Seeker_id = $id";
$userResult = $conn->query($sql);
 if ($userResult->num_rows > 0)
 {$userData=$userResult->fetch_assoc();
$recommendedJobs= recommendJobsFromDatabase($conn, $userData);
```

It executes an SQL query to fetch user data from the seekerresume table based on a specified $id. It checks if user data exists in the database. If user data is found, it retrieves the user's profile information. It then calls the recommendJobs function to generate job recommendations for the user based on their skills.

```
$userSkills = explode(',', $userData['skill']);
```

This line of code splits a comma-separated string of user skills into an array of individual skills.

```
$sql = "SELECT j.job_id, j.title, j.location, j.skills, j.salary, e.employer_id, e.Fullname_E
FROM job_postings AS j JOIN employerlogin AS e ON j.employer_id = e.employer_id";
$result = $conn->query($sql);
```

```
if ($result->num_rows > 0) {
while ($job = $result->fetch_assoc()) {
```

It constructs an SQL query to retrieve job-related information, including job ID, title, location, skills required, salary, employer ID, and employer name. It executes the query using the database connection represented by $conn and stores the result in the $result variable. It checks if there are any rows in the query result. If rows are found, it enters a loop to fetch each job's details as an associative array called $job.

```
$jobSkills = explode(', ', $job['skills']);
 $matchingSkills = array_intersect($userSkills, $jobSkills);
```

It takes the job's required skills, which are stored as a comma-separated string in the $job['skills'] field, and splits them into an array called $jobSkills. Each skill becomes a separate element in this array. Then the code compares the user's skills with the skills required for the job in the $jobSkills array. array_intersect function is used to find the common skills between the two arrays. As a result, $matchingSkills contains the skills that both the user and the job share, indicating how well the user's skills match the job's requirements. These matching skills are valuable for making job recommendations that closely align with the user's skillset.

```
if (!empty($matchingSkills)) {
$job['matching_skills'] = implode(',', $matchingSkills);
$job['matching_skill_count'] = count($matchingSkills);
```

Code checks if an array $matchingSkills is not empty. If it's not empty, it creates a comma-separated string of the skills and stores it in $job['matching_skills']. It also counts the number of skills in the array and stores that count in $job['matching_skill_count'].

4.2 Testing

4.2.1 Test case for Unit Testing

Table 4.1: Test Table for Signup and Login

Objectives	To test user registration and login
Test I	
Action	User credential was inserted and login was initiated
Expected Result	Could not login without registration.
Actual Result	Could not login without registration.
Test II	
Action	Registration was initiated by filling registration form
Expected Result	Signup Successful
Actual Result	Signup Successful
Test III	
Action	User credentials were inserted for login
Expected Result	User can now fill CVs and get the personalized recommendation
Actual Result	User can now fill CVs and get the personalized recommendation
Conclusion	Test Successful

Table 4.2: Test Table for Resume

Objectives	To test resume form
Test I	
Action	Filled up the all-user’s information on resume form and submitted.
Expected Result	Form inserted and now you can see the recommended jobs as per your skills.

Actual Result	You can see the recommended jobs as per your skills.
Test II	
Action	Trying to submit form without filling all the fields on the form.
Expected Result	Asks to fill that particular field, no submission made.
Conclusion	Test Successful

Table 4.3: Test Table for Job Posting

Objectives	To test job posting form on employer's end
Test I	
Action	Filled up the job information on job posting form and submitted.
Expected Result	Form inserted and you can see your posted job on left hand side.
Actual Result	You can see your posted job on left hand side.
Conclusion	Successful

Table 4.4: Test Table for Salary Prediction

Objectives	To test the prediction of salary.
Test I	
Action	Filled the fields with skills and year of experience
Expected Result	Predicted salary is shown right below the form.
Actual Result	Should predict salary based on the skills and year of experience
Conclusion	Successful

Chapter 5: Conclusion and Future Recommendation

5.1 Lesson Learnt/ Outcome

The primary objective of undertaking this project is to gain valuable insights into the effective implementation of an appropriate algorithm that aligns well with the project's objectives. This initiative has provided us with a valuable opportunity to enhance our skills and knowledge, leading to a meaningful and enriching learning experience.

5.2 Conclusion

Job portals are important platforms that help job seekers find the right job opportunities and connect with potential employers. With this project, job seekers can easily create accounts and log in to the application. They can then explore and apply for jobs that match their skills. The system uses clever algorithms to suggest jobs based on skills. This makes it easier to find suitable jobs. Employers also benefit by using the system to find qualified candidates for their job openings. Job seekers can add, update, and remove their resumes, and employers can post job listings, review resumes, and communicate with potential hires. Once employers have filled a position, they can remove the job listing from the portal.

5.3 Future Recommendation

The Future scope includes the integration of chatting application where job seeker can directly contact to the employers through the portal. This system can be made more user friendly with better responsiveness.

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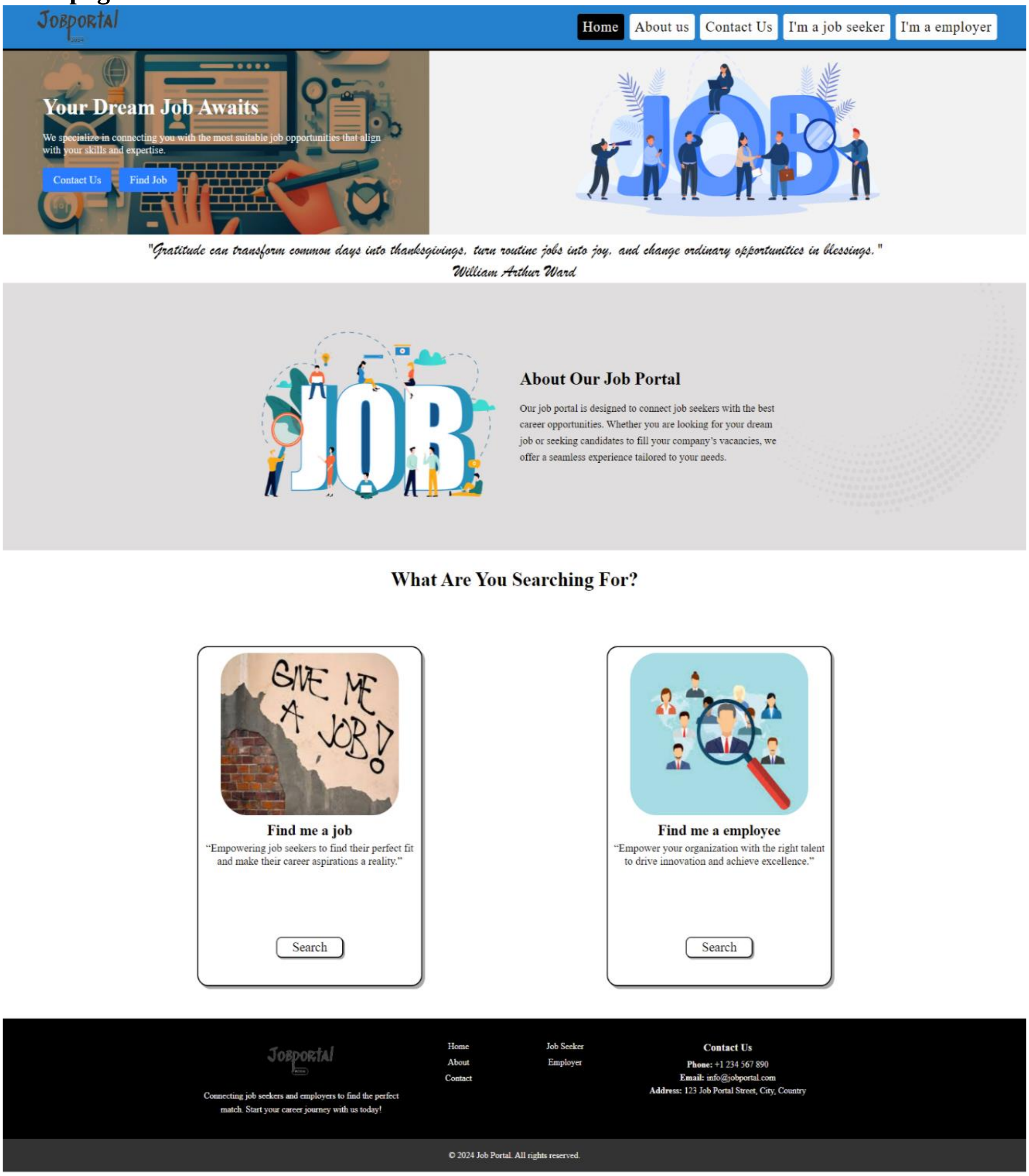
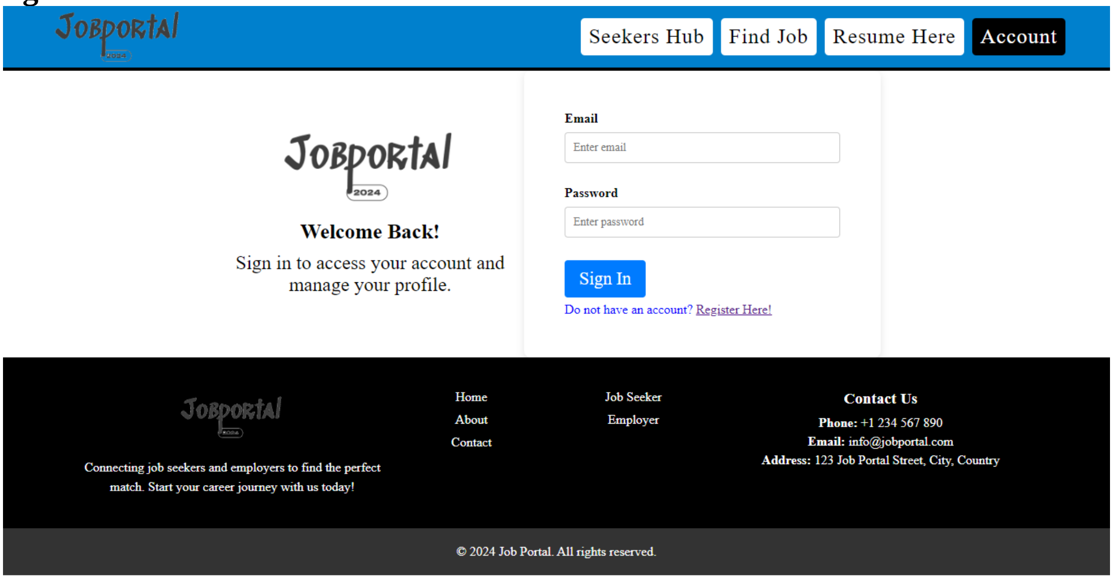
Appendices

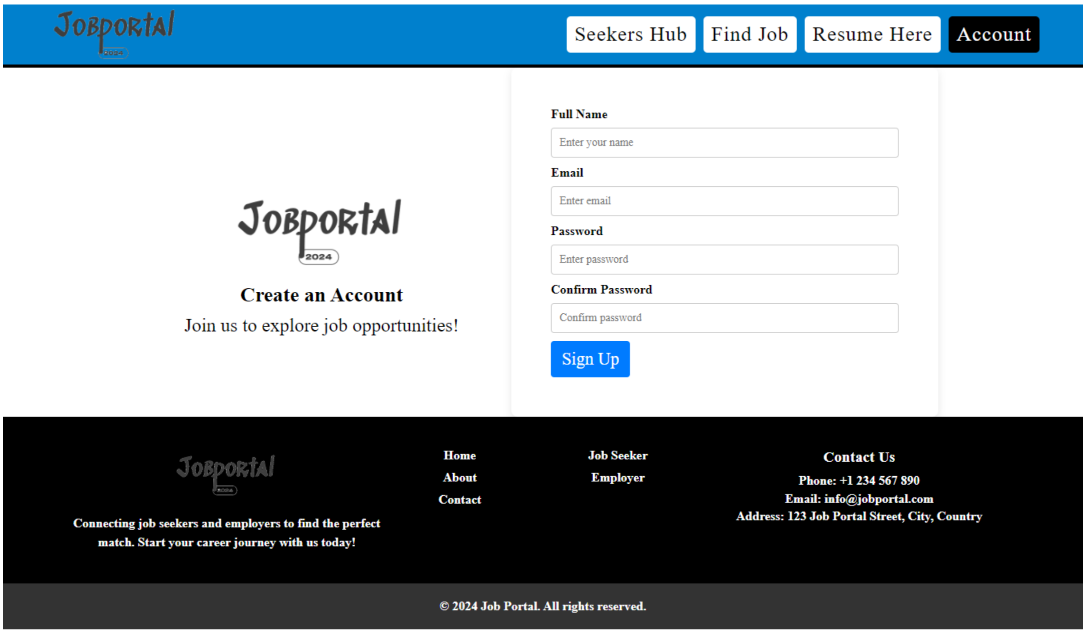
Homepage



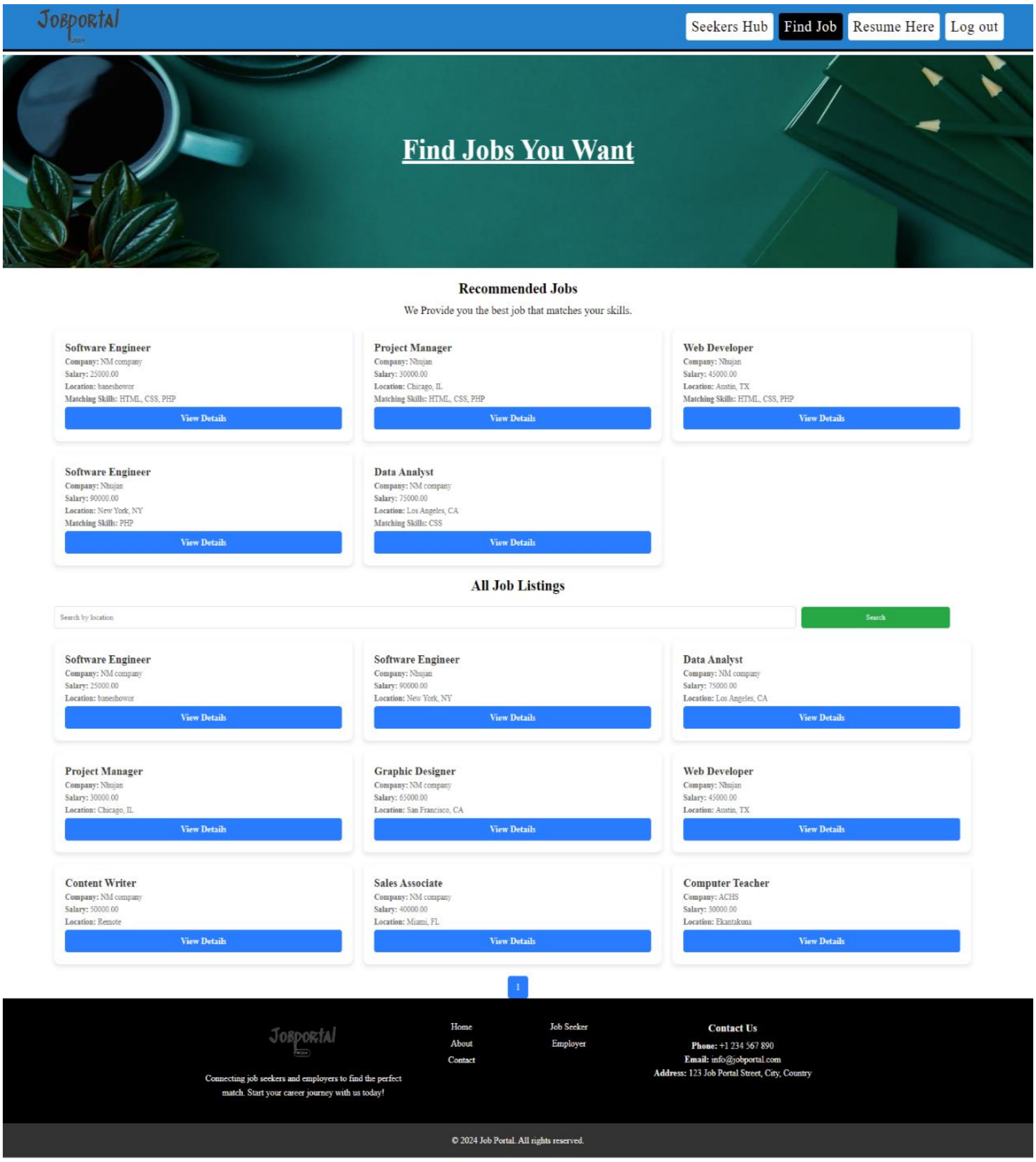
Sign In



Sign up



Job List


Seekers Hub
Find Job
Resume Here
Log out
Find Jobs You Want
Recommended Jobs
We Provide you the best job that matches your skills.
Software Engineer
View Details
Project Manager
View Details
Web Developer
View Details
Software Engineer
View Details
Data Analyst
View Details
All Job Listings
Search
Software Engineer
View Details
Software Engineer
View Details
Data Analyst
View Details
Project Manager
View Details
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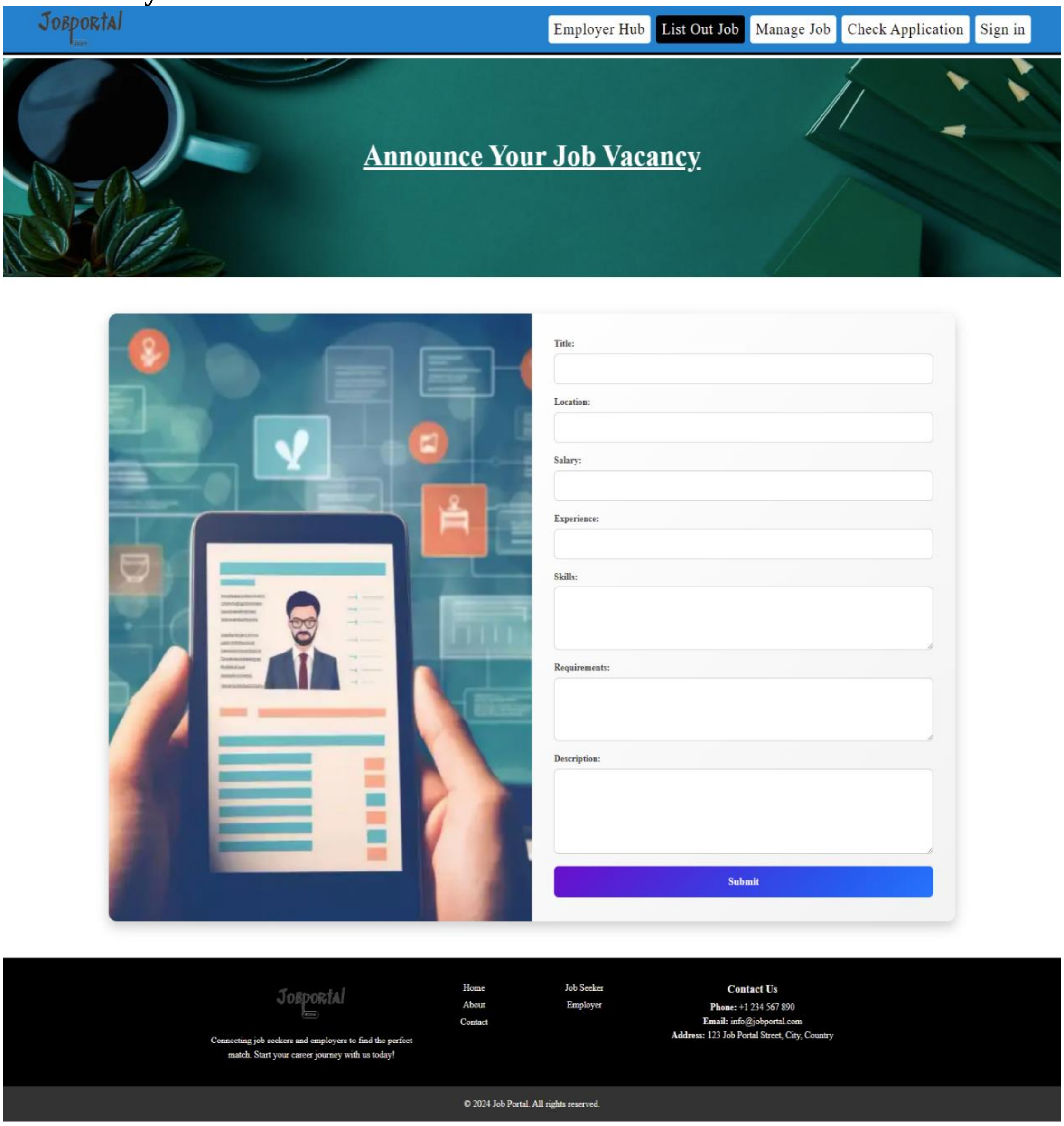
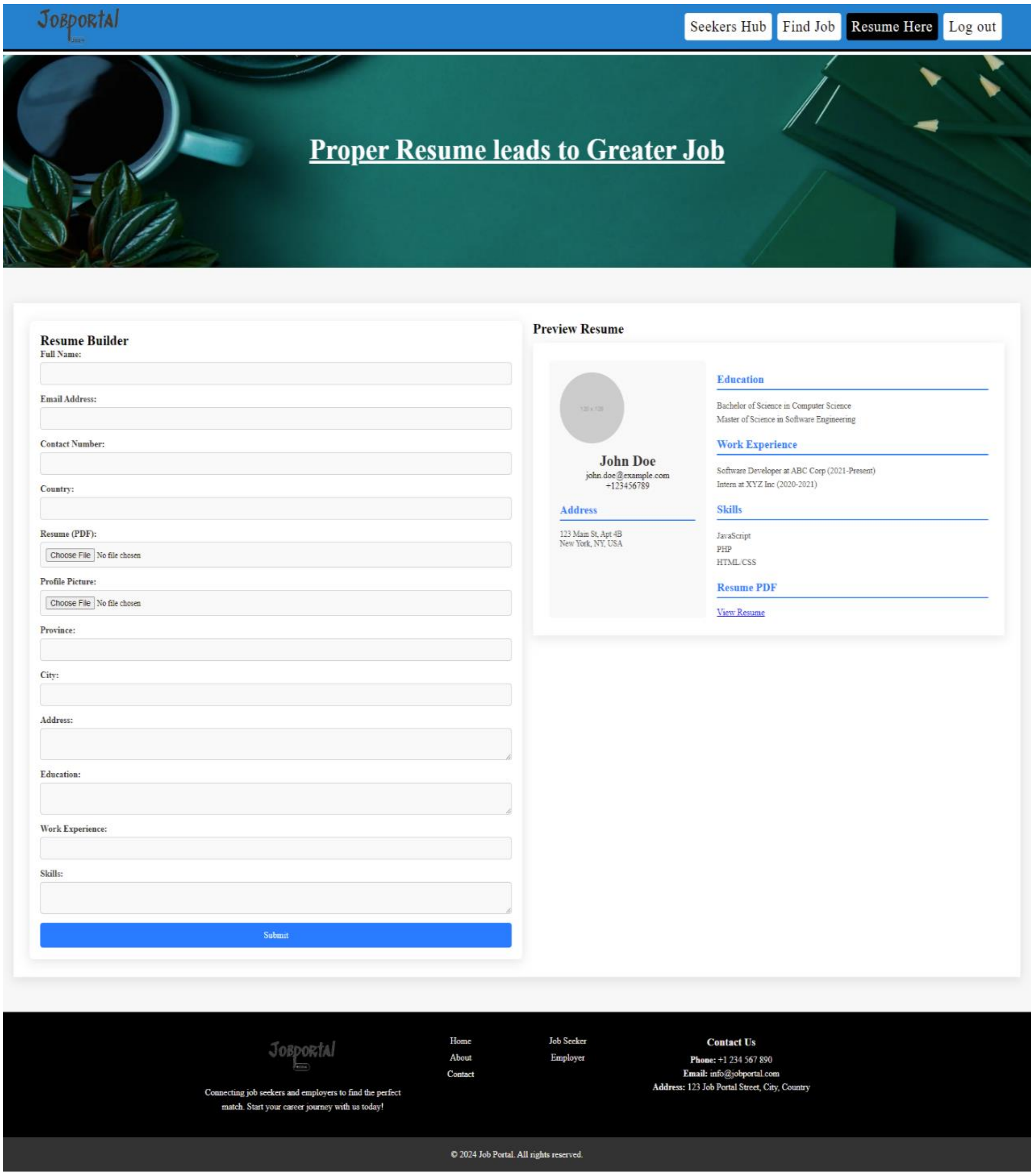
Job Vacancy Form



Resume Form



Seekers Hub
Find Job
Resume Here
Log out
Proper Resume leads to Greater Job
Resume Builder
Full Name:
Email Address:
Contact Number:
Country:
Resume (PDF):
Choose File No file chosen
Profile Picture:
Choose File No file chosen
Province:
City:
Address:
Education:
Work Experience:
Skills:
Submit
Preview Resume
John Doe
john.doe@example.com
+123456789
Address
123 Main St, Apt 4B
New York, NY, USA
Education
Bachelor of Science in Computer Science
Master of Science in Software Engineering
Work Experience
Software Developer at ABC Corp (2021-Present)
Intern at XYZ Inc (2020-2021)
Skills
JavaScript
PHP
HTML/CSS
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